

(1.1) DokaŹte, Źe v libovolnm komplexnm vektorovm prostoru $\mathbf{V}$ se skalrnm souinem $\langle \cdot, \cdot \rangle$ plat (pro libovoln $\mathbf{u}, \mathbf{v} \in V$)

$$\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2), \quad (1)$$

$$\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2), \quad (2)$$

kde $\operatorname{Re}(x)$ zna relnou a $\operatorname{Im}(x)$ imaginrn ast komplexnho sla x . (Pro inspiraci vizte tvrzen 8.32 ve skriptech.)

Nejprve si vypšme nkter znm rovnosti:

$$1. \|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} \Leftrightarrow \|\mathbf{u}\|^2 = \langle \mathbf{u}, \mathbf{u} \rangle$$

$$2. \langle \mathbf{u} + \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle$$

$$3. \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$$

$$4. \langle c \cdot \mathbf{u}, \mathbf{v} \rangle = \bar{c} \cdot \langle \mathbf{u}, \mathbf{v} \rangle$$

$$5. \langle \mathbf{u}, c \cdot \mathbf{v} \rangle = c \cdot \langle \mathbf{u}, \mathbf{v} \rangle$$

$$6. \langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$$

kde $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbf{V}$, $c \in \mathbb{C}$.

Podvejme se nejprve na rovnost (1) ze zadn

$$\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$$

Nyn rozepšme pomocí znmch rovnost vše leny na prav stran rovnosti. Tedy plat:

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} + \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + \langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} \end{aligned}$$

Vezmme si nyn libovoln $z \in \mathbb{C}$, $z = a + ib$, $a, b \in \mathbb{R}$. Plat: $z + \bar{z} = a + ib + a - ib = 2a$. Tedy $\forall z \in \mathbb{C}$ plat: $z + \bar{z} = 2\operatorname{Re}(z)$. V našem ppd plat, Źe $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{C}$, tedy plat: $\langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} = 2\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle)$ a mžeme tedy pst (z rovnosti vše), Źe:

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle)$$

. A tedy obdobn mžeme odvodit, Źe

$$\begin{aligned} \|\mathbf{u} - \mathbf{v}\|^2 &= \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} - \mathbf{v} \rangle + \langle -\mathbf{v}, \mathbf{u} - \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - \langle \mathbf{u}, \mathbf{v} \rangle - \overline{\langle \mathbf{u}, \mathbf{v} \rangle} \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) \end{aligned}$$

Mme tedy rovnice:

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) \quad (3)$$

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) \quad (4)$$

Odetme nyn rovnici (4) od rovnice (3), dostaneme:

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) - \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 + 2\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) \\ &= 4\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) \end{aligned}$$

A tedy vynásobením této rovnosti hodnotou $\frac{1}{4}$ dostaneme:

$$\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$$

což je přesně rovnice (1) ze zadání.

Přejdeme nyní k rovnosti (2):

$$\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2)$$

Opět můžeme psát

$$\begin{aligned} \|\mathbf{u} - i\mathbf{v}\|^2 &= \langle \mathbf{u} - i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} - i\mathbf{v} \rangle + \langle -i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle - i\langle \mathbf{u}, \mathbf{v} \rangle + i\langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - i(\langle \mathbf{u}, \mathbf{v} \rangle - \overline{\langle \mathbf{u}, \mathbf{v} \rangle}) \\ \|\mathbf{u} + i\mathbf{v}\|^2 &= \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} + i\mathbf{v} \rangle + \langle i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + i\langle \mathbf{u}, \mathbf{v} \rangle - i\langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + i(\langle \mathbf{u}, \mathbf{v} \rangle - \overline{\langle \mathbf{u}, \mathbf{v} \rangle}) \end{aligned}$$

Vezměme si nyní opět libovolné $z \in \mathbb{C}$, $z = a + ib$, $a, b \in \mathbb{R}$. Platí $z - \bar{z} = a + ib - (a - ib) = 2ib = 2i\operatorname{Im}(z)$.

V našem případě tedy platí: $\langle \mathbf{u}, \mathbf{v} \rangle - \overline{\langle \mathbf{u}, \mathbf{v} \rangle} = 2i\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle)$. Nyní tedy máme následující rovnice:

$$\|\mathbf{u} - i\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - i(2i\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle)) \quad (5)$$

$$\|\mathbf{u} + i\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + i(2i\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle)) \quad (6)$$

Odečtíme nyní rovnici (6) od rovnice (5):

$$\begin{aligned} \|\mathbf{u} - i\mathbf{v}\|^2 - (\|\mathbf{u} + i\mathbf{v}\|^2) &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2i(2i\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle)) - \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 \\ &= 4\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) \end{aligned}$$

Po vynásobení této rovnice hodnotou $\frac{1}{4}$ dosaneme:

$$\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2),$$

což je přesně rovnost (2) ze zadání. Tímto je tedy důkaz rovností ze zadání proveden.